MILITARY SPECIFICATION

BUR, BONE SURGERY (Tungsten Carbide), GENERAL REQUIREMENTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers general requirements for tungsten carbide, bone surgery burs used in DoD Deployable Medical Systems (DMS) and is intended to be used in conjunction with the detailed military specification sheets or procurement documents covering such additional requirements as are applicable to each individual bur.

1.2 Classification.

1.2.1 Head shapes and shank lengths. The surgical burs shall be of the following head shapes and shank lengths as specified in the applicable specification sheets (see 6.2):

Shape:

A. Round Cutting
B. Side Cutting, Straight
C. Side Cutting, Tapered
D. Oval Cutting
E. Barrel Cutting

Shank Length:

1. Short
2. Medium
3. Long
4. Extra Long

AMSC N/A FSC 6515

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

MILITARY

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-P-116	Methods of Preservation.

STANDARDS

FEDERAL

FED-STD-151	Metals: Test Methods.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and requlations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. All burs shall conform to the requirements specified herein and the military specification sheets applicable to the individual bur specified.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable Military Specification Sheets. In the event of any conflict between requirements sheets, the latter shall govern.

3.2 Design and construction. The burs shall be surgical, tungsten carbide burs suitable for use in bone surgery when used in the Pneumatic Bone Drill, National Stock Number (NSN) 6515-01-116-8197. The bur head shall have properly and uniformly sharpened cutting edges. The style, design, and dimensions of the burs shall be as specified in the detail specification sheets for the applicable NSN (see 6.2).

3.3 Runout. The total indicated runout of the burs shall not exceed .002 inch.

3.4 Finish. The bur head shall have a finely honed finish with sharpened blades. The neck and shank of the burs shall be smooth, and any safeline present on the shank shall have an etched finish.

3.5 Autoclaving. The burs shall show no corrosion when tested as specified in 4.4.2.

3.6 Workmanship. The burs shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of burs of the same size, manufactured under essentially the same conditions. Each unit package of burs shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

4.2 Quality conformance inspection. The examinations required to assure conformance of those burs or lots of burs to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3). The tests required for acceptance are specified in table II of 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptance quality levels (AQLs) shall be as shown in table I. The unit of product for examination purposes shall be in the individual bur.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

TABLE I. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.5
Major B	S-1	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be as stated in the table.

TABLE II. Sampling criteria for tests

Characteristics	Requirements	Test procedures	Inspection levels
Chemical Composition	3.1	4.4.1	see 4.2.3
Autoclaving	3.7	4.4.2	S-1

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

4.3 Examination. The burs shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the following classification of defects listed in table III. Examination shall not be restricted to the classified defects shown in table III.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects

Category	Defect
Critical:	None defined.
Major A:	
101	Style and design not as specified.
102	Bur size or head shape not as specified.
103	Total indicated runout exceeds 0.002 inch.
104	Head diameter exceeds specification requirements at maximum cross section.
105	Etched safeline missing.
Major B:	
141	Operative end and shank dimensions not as specified (except for 104).
Minor:	
201	Cutoff end on shank not as shown in applicable figure.
202	Burs not free from defects such as chipped blades, cracks, or pits.
203	Finish on bur not as specified.
204	Overall length of bur not within specification limits.
205	Instructions for use inaccurate, inadequate, or missing.

4.3.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

4.3.2 Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360 degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.3.3 Visual examination. Examination for possible bur defects not perceptible to the unaided eye shall be made using a 30X magnification, unless otherwise specified in individual specification sheet (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples selected as specified in 4.2.3 shall be analyzed in accordance with Method 111 or 112 of Fed. Std. No. 151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.2 Autoclaving. The test burs shall be cleaned in accordance with the instructions supplied by the manufacturer. The burs shall then be autoclaved at 270 ± 5 degrees Fahrenheit (F.) and 27 ± 1 pounds per square inch pressure for a minimum of 20 minutes. After the burs are removed from the autoclave, the entire surface area of the bur shall be examined for evidence of corrosion.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1. Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each bur shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each bur shall be wrapped in neutral tissue or other suitable non-corrosive material. Each bur shall be further packaged in accordance with MIL-P-116, method IC-3.

5.1.1.4 Intermediate package. Twelve burs of one shape and length only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for burs shall conform to the MIL-STD-794 requirements for this level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) containing burs of one shape and length only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the applicable box specification.

5.2.2 Level B. Twenty-four intermediate packages (288 units) containing burs of one shape and length only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for burs shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Burs packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Burs packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. Tungsten carbide, bone surgery burs covered by this specification are intended for use in medical war readiness assemblages and DoD Deployable Medical Systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet.
c. NSN (as applicable).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Subject term (key word) listing:

DEPMEDS
Surgical Burs
Tungsten Carbide Burs

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6515-3794

Location: NCR Disk 5

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvement. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 6, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments was received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4968 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
Attn: DPSC-RSTS
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-42017	BUR, BONE SURGERY (Tungsten Carbide) General Requirements for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 20261O-21T20:10Z
Check the source to verify that this is the current version before use.